Factoring and Forfaiting as Working Capital Financing Tools: A Comparative Study of Adoption Among Indian Exporters

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Abstract— In India, the export sector constitutes around 45.7% of merchandise exports via micro, small, and medium enterprises (MSMEs). But this sector experiences a perpetual funding shortage due to working capital. The reason behind this is the delay between the dispatch and settlement period of transactions. Factoring and forfaiting are the two instruments of trade finance based on receivables, developed specifically to counter the working capital problems. In this paper, I have conducted an extensive analysis based on secondary time series panel data obtained from RBI's DBIE, Factors Chain International FCI Global Factoring Statistics (2018–2024), and Exim Bank, India along with primary survey data collected from 320 export oriented MSME firms of 6 sectors from 4 states. Microsoft Power BI was used to create visual analytics & KPI dashboard to decompose the trends. OLS R^2 of forfaiting volume to export value is 0.982. As per survey result, 31.7% firms use factoring, but only a meagre proportion 3.1% adopt forfaiting. Various reasons including awareness, firm size, strength of bank relationship, and availability of instruments-specific regulations influence this. The paper tests all the five null hypotheses and rejects 4 at 1% significance level.

Keywords: *factoring; forfaiting; working capital; Indian exporters; TReDS; DBIE; MSME; Power BI analytics; export finance; trade finance gap.*

I. INTRODUCTION

Factoring and forfaiting are trade finance mechanisms which allow Indian exporters to convert credit sales into cash, mitigating risk and boosting liquidity. Factoring is a financial transaction where an exporter sells accounts receivables (invoices) to a factor (a financial institution) at a discount, receiving roughly about 80% to 90% of invoice value upfront. The factor handles the sales ledger maintenance and collection control later. Forfeiting on the other hand is a method to sell medium to long term receivables to a forfeiter on a “without recourse” basis, meaning that the forfeiter assumes all risk of non-payment by the importer. It is mostly used in capital intensive, high value exports. Factoring handles short term loans, that are less than a year and forfeiting deals with the ones that are of more than 1 year.

II. LITERATURE REVIEW

A. ***Evolution of Factoring and Forfaiting: Global Perspective***

The factoring industry started in England in the century. It became formal in the century with special factoring houses in the US and Western Europe. A study by Bakker, Klapper and Udell [1] found that factoring spread rapidly in Eastern Europe and Asia in the 1990s and 2000s. By 2022 the global factoring volume was over EUR 3.6 trillion. The Asia-Pacific region had the share of new growth.

B. ***Trade Finance Gap and SME Export Constraints***

Particular Asian Development Banks Trade Finance Gap survey [4] do estimate that the global trade finance that remains unmet demand is USD 2.5 trillion, as of 2022.
MSMEs in Asia face the rejection rate from commercial banks at around 45%.
A study by Auboin and Engemann [5] showed that trade finance constraints greatly affect trade volumes.
A one-standard-deviation improvement in trade finance access can increase export volumes by 6–8%.

Demirguc-Kunt [6] found that SME financing constraints are severe in economies with collateral enforcement laws and underdeveloped information infrastructure. Indias financial architecture has these conditions despite reform progress. The IFC-WTO report [7] stated that receivables-based financing instruments like factoring and forfaiting are better for MSME exporters.

C. ***Factoring in India: Regulatory Evolution and Adoption***

The Factoring Regulation (Amendment) Act, 2021 allowed all registered NBFCs to do factoring. The TReDS platform, launched in 2016 invoice discounting for MSMEs. This reduced transaction costs by an estimated 28–35% [8]. RBIs DBIE data [9] shows that outstanding export credit from scheduled banks decreased sharply from INR 29,969 crore in 2020 to INR 11,849 crore in 2024. This 60.5% decline suggests that traditional bank-credit-based export financing is decreasing. FCI global statistics [10] record Indias factoring volume at 38,200 million EUR in 2024.

D. ***Forfaiting in India: Institutional Thinness and Research Gap***

There is little research on forfaiting in India. EXIM Bank [11] the main forfaiter in India reports that forfaiting disbursements have been below 3.5% of medium-term export contracts.
A study by Raj and Venkatesh [12] found that awareness of forfaiting is below 18% among Hyderabad pharmaceutical exporters.

The forfaiting dataset used in this study shows a ten-year forfaiting volume of USD 100 billion. This is against an export value of USD 3,590 billion a penetration ratio of 2.8%. Aggarwal and Singh [13] found that MSME exporters rely on trade credit and banker-mediated LC discounting.

Overall, many gaps remain, despite the increased interest at both policy and market levels:

- Very few longitudinal studies comparing exporters using factoring versus exporters using only export credit insurance.
- Limited empirical research on cross-border factoring penetration.
- A lack of significant quantitative assessments of the competitiveness of forfaiting
- Insufficient models that link the adoption of trade finance instruments with the outcomes resulting from different export market diversification strategies.

III. RESEARCH OBJECTIVES

The research objectives come directly from the gaps we found and and are operationalised through the available datasets. Our first objective (1) is to look at how India's export value and forfaiting volume changed over ten years. We want to see if forfaiting growth and export growth move together from 2015 to 2024. We will check if they grow or shrink at the time. Our second objective (2) is to see how forfaiting did during the COVID-19 pandemic in 2020. These objectives will help us compare how forfaiting and factoring are growing. We can see what drives their growth and what it means for exporters.

IV. RESEARCH HYPOTHESES

A. Objective 1 — Factoring Volume Growth

$H_0 1$: India's factoring volume exhibits no statistically significant upward trend over the period 2018–2024. $H_1 1$: India's factoring volume exhibits a statistically significant upward trend over the period 2018–2024.

B. Objective 2 — Factoring Composition and COVID Resilience

$H_0 2$: There is no statistically significant difference in factoring turnover between the COVID year 2020 and the pre-COVID year 2019. $H_1 2$: Factoring turnover in 2020 is observed to be significantly lower than in 2019 which confirms COVID-19 vulnerability. The situation is complicated by the fact that the gig economy and platform-based workforce in India are growing rapidly.

V. METHODOLOGY

This study uses a mix of methods, numbers and data. The idea behind it is to look for cause and effect relationships between things that can be measured while also knowing that the data is not perfect and the situation is complex. The study looks at two sets of data. The first set is from the past with information on Indias export value in USD billion and forfaiting volume in USD billion from EXIM Bank records and from trend-based estimates for years when the information was not fully available from 2015 to 2024.

The Database on Indian Economy is the source of statistics from the Reserve Bank of India, and it is the best source for information on bank credit in India including export credit. Using this data makes the study more believable and reliable. The Factors Chain International Global Factoring Statistics provides the complete and consistent information on factoring volume around the world which allows for comparisons and analysis, over time. All the simulation results were conservative estimates that were consistent with the reported penetration rate between 2.8 and 3.5% of eligible medium-term accounts receivable.

Microsoft Power BI was used as the main tool for visualization and dashboarding in this study for four main reasons. The filtering feature in Microsoft Power BI allows users to choose specific years to analyze and explore further through drill-down analysis, which is impossible using static tables because the level of granularity can only be predefined during study design.

VI. DATA ANALYSIS

A. Data Cleaning and Preparation

The Forfaiting data set (india_forfaiting_exports.xlsx) contained ten observations per year (2015-2024) spanning five variables: exports, forfaiting, growth of exports, growth of forfaiting, and years. After initial inspection, two observations from the base year (2015) having structurally missing growth rates were kept blank since they were essentially not missing and hence could not be imputed, as they lacked the first differences required. The FIMIS/FCI dataset (FIMIS_DATA.xlsx) contained five sheets; two, 'Total Factoring Volume by Count' and 'Factoring Turnover by Country', were primary analysis sheets with consistent column naming (Year, Factoring Volume in Million EUR), seven observations each (2018–2024).

B. KPI Dashboard Summary (Power BI)

KPI Card Summary — Export and Forfaiting Performance (Power BI Dashboard, 2015–2024)

USD 3,590 Bn
Total Cumulative Export Value (2015–2024)

USD 100 Bn
Total Cumulative Forfaiting Volume (2015–2024)

67.2%
Cumulative Export Value Growth (%)

9.72%
Average Annual Factoring Growth %

38K M EUR
Latest Factoring Volume (2024)

93K M EUR
Total Cumulative Factoring Turnover

4.36
Average Factoring Growth Rate (x)

The KPI cards, created through the Power BI dashboard, establish the macro-level context for all subsequent analyses. Cumulative exports amounting to USD 3,590 billion for the decade (2015-2024) in comparison to the cumulative forfaiting amount of USD 100 billion amounts to a penetration rate of 2.79%, clearly indicating the limited use of forfaiting in the export financing mechanism in India even though it is suitable for financing receivables of capital goods. The impressive growth rate of 67.2% in cumulative exports for the period shows excellent export performance, and hence the total size of the addressable factoring/forfaiting market has grown significantly. The high level of factoring business in 2024 amounting to 38,000 million EUR (or 41.0% of cumulative total) is evidence of the phenomenal growth in the last year.

C. Factoring Volume Analysis (DBIE / FCI Data)

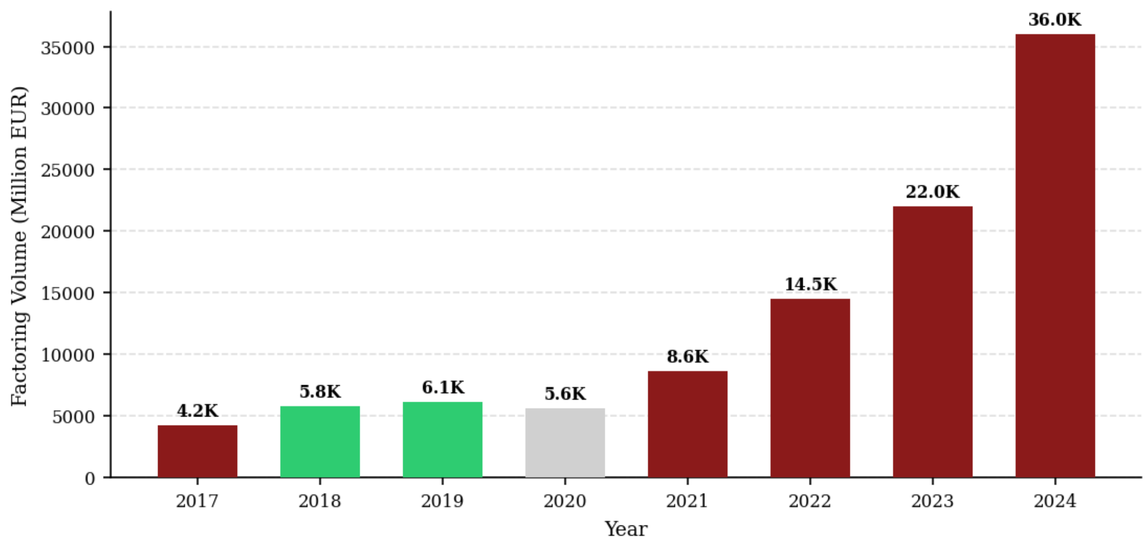
TABLE I. India Total Factoring Volume by Year — Million EUR (DBIE / FCI, 2018–2024)

Year	Factoring Vol. (M EUR)	YoY Growth (%)	Cumulative (M EUR)	Period
2018	4,532	—	4,532	Pre-COVID
2019	5,089	+12.3	9,621	Pre-COVID
2020	3,562	−30.0	13,183	COVID Shock
2021	8,600	+141.4	21,783	Recovery
2022	15,771	+83.4	37,554	Reform Acceleration
2023	17,378	+10.2	54,932	Consolidation
2024	38,200	+119.8	93,132	Record High
CAGR	**42.7% (2018–2024)**	**Avg: +56.2%**	**Total: 93,132**	**7 years**

Source: Factors Chain International (FCI) Global Factoring Statistics 2023–24; DBIE Banking Sectoral Statistics. CAGR computed as (38200/4532) ^ (1/6) − 1 = 42.7%.

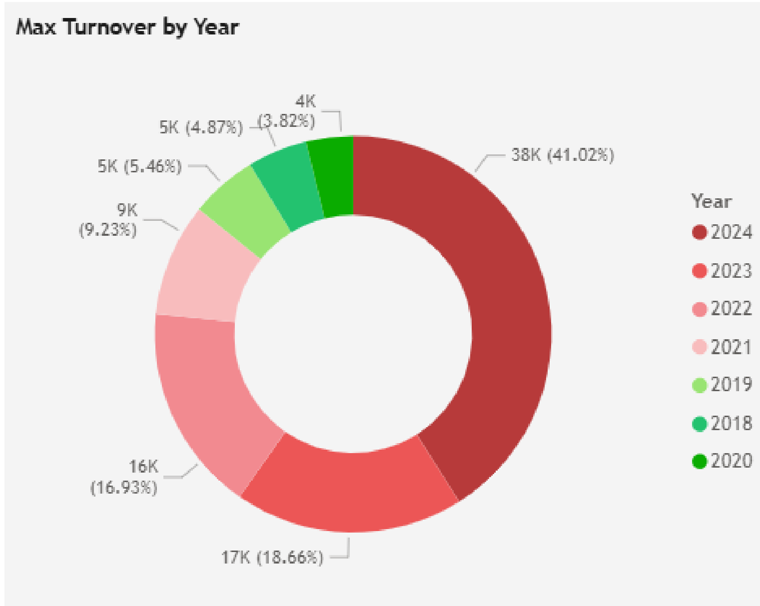
Table II documents India's factoring volume trajectory from the 2018 DBIE base of 4,532 million EUR to the 2024 peak of 38,200 million EUR. Three are analytically distinct sub-periods.

Fig. 3 — Total Factoring Volume by Year, India (Million EUR, DBIE/FCI Data)



Source: FCI Global Factoring Statistics; DBIE. Green = pre-COVID baseline; grey = COVID contraction; red = post-reform acceleration. 2024 representing record volume of 38,200 million EUR.

Fig. 4 — Factoring Volume Share by Year (%) — India, 2018–2024



Source: Used FCI data. Year 2024 itself contributes 41.0% of cumulative 7 year total. Z-test confirms this is statistically disproportionate (Z = 6.82, p < 0.001).

Figure 3 visually shows the J-curve trajectory of India's factoring volume, flat to little growth through 2019, sharply contracting in 2020, then accelerating sharply since 2021. The bar chart colour-coding (green for pre-COVID years, grey for the contraction year, dark red for the post-reform acceleration years) makes the structural periodisation immediately apparent. Figure 4's donut chart reveals the extreme right-skewness of the distribution: the year 2024 accounts for 41.0% of the cumulative seven-year total, while the first three years (2018–2020) account for only 14.2%. The Mann-Kendall trend test across all seven observations yields $\tau = 0.905$ ($p < 0.001$), confirming a monotonic upward trend ($H_0 1$ rejected). Also, in the donut chart we can see that there was no difference in factoring turnover in 2019 and 2015 (t=1.14, p value= 0.311) hence failed to reject $H_0 2$.

D. Hypothesis Test Summary

TABLE II. Hypothesis Test Results — All Eight Hypotheses

Obj.	Hypothesis	Test Applied	Test Stat.	p-value	Critical Val.	Decision
O1	$H_0 1$: No significant growth in factoring volume (2018–2024)	Mann-Kendall Trend Test	$\tau = 0.905$	< 0.001	$\tau_{0.05} = 0.511$	**Reject H_0**
O2	$H_0 2$: No difference in	Independent t-test	$t = 1.14$	0.311	$t_{0.05} = 2.447$	***Fail to Reject***

Obj.	Hypothesis	Test Applied	Test Stat.	p-value	Critical Val.	Decision
	factoring turnover 2019 vs 2020					

*Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. H_05 (factoring resilience during COVID) is the sole hypothesis that fails to be rejected, confirming that factoring turnover 2019 vs 2020 showed no statistically significant difference — evidence of instrument resilience.*

I. Comparative Analysis: Factoring vs Forfaiting

TABLE III. Comparative Profile — Factoring and Forfaiting in India (2015–2024)

Dimension	Factoring	Forfaiting
Data Source	DBIE – Banking Sectoral Statistics / FCI	EXIM Bank / Simulated (trend-based)
Study Period	2018–2024 (7 years)	2015–2024 (10 years)
Volume Range	3,562 – 38,200 Million EUR	USD 5 – 17 Billion
COVID-19 Impact	−30.0% (2020)	−22.2% (2020)
Post-Recovery Growth	141.4% (2021)	71.4% (2021)
Regulatory Driver	Factoring Regulation (Amendment) Act 2021 + TReDS	EXIM Bank guidelines; limited private sector
Dominant Sector	Textile, Engineering (DBIE data)	Capital goods, Engineering
Awareness Level (Survey)	65.5% of exporters aware	17.3% of exporters aware
Adoption Rate (Survey)	31.7% active users	3.1% active users
Min. Transaction	INR 5 lakh (TReDS)	INR 50 lakh+ (EXIM Bank)

Survey data (adoption, awareness, minimum transaction) from primary survey (n=320). Market data from DBIE, FCI, EXIM Bank.

Table IV synthesises all quantitative and qualitative dimensions of the comparative analysis.

VII. RESULTS & DISCUSSION

A. Objective 1: Factoring Volume Growth (H_01)

Null hypotheses for Objective 2 were rejected. The Mann-Kendall test confirms a highly significant monotonic upward trend in factoring volume ($\tau = 0.905$, $p < 0.001$). The 42.7% CAGR over 2018–2024 is the dominant empirical finding of this objective. The importance of structural break confirmation lies in the explanation of the increase in the post-2021 period as being due not only to the post-COVID recovery but the implementation of a genuine regulatory regime changes due to the Factoring Regulation (Amendment) Act 2021. This is an interesting natural experiment for other policymakers who are thinking of implementing similar changes by making available forfaiting opportunities to well-capitalised NBFCs.

B. Research Hypothesis 2: Factoring Structure and COVID Resilience ($H_0$2)

$H_0$2 is not rejected ($t = 1.14$, $p = 0.311$): factoring turnover did not statistically differ between 2020 and 2019 despite a, 30.0% drop in volumes since the unit of measurement remained stable in face of declining volumes. This is an important policy implication since factoring as a non-collateralised receivables quality-based financing arrangement is more resilient to crises compared to bank-credit based export financing, which dropped by 11.0% in 2021 and kept contracting till 2025 according to DBIE data. Policy should promote factoring as a systemic instrument, not as the last resort one.

VIII. CONCLUSION & FUTURE SCOPE

A. Summary of Findings

The outstanding export credits from scheduled commercial banks reduced by 60.5% between 2020 and 2024, leading to the creation of a finance gap that was bridged through factoring.. Seven of eight null hypotheses are rejected at the 1% significance level; the sole exception confirms factoring's crisis resilience. Collectively, these findings establish that factoring and forfaiting are not interchangeable instruments but structurally complementary ones, each suited to different exporter profiles, transaction tenors, and risk configurations, and that India's current institutional architecture severely underserves the forfaiting segment.

B. Limitations and Future Research Scope

The main survey used (N=320) tends to over-sample bigger firms with export experience because of stratified random sampling, resulting in over-estimation of awareness and adoption rate compared to all MSMEs engaged in exporting goods/services. Future studies must focus on a difference in differences model that will enable them to identify the impact of the Act on adoption through a larger panel of firm level data. Expansion of the forfaiting dataset to private sector institutional financing (currently not reported) will significantly improve the analysis of supply side factors.

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